

Ideation Phase

Define the Problem Statements

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized

recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Customer Problem Statement — Three Primary Personas

Problem Statement (PS)	I am (Customer)	I'm trying to...	But...	Because...	Which makes me feel...
PS-1 (Priya — Skincare Consumer)	A skincare consumer with sensitive skin and a limited budget	Choose well-rated cosmetics products suited to my skin type	I don't have a single place to compare price against rating and skin-type suitability	Product listings and reviews are scattered across retailer sites and not presented visually	Frustrated, and forced to rely on guesswork and influencer recommendations
PS-2 (Meera — Retail Category Manager)	A retail category manager deciding which brands and categories to stock	Identify which cosmetics categories and brands are worth stocking and how to price them	I only have access to static spreadsheet exports from suppliers	There is no interactive, filterable view of pricing and rating patterns across the catalog	Uncertain, and unable to quickly justify stocking decisions with evidence
PS-3 (Arjun — Brand Marketing Analyst)	A brand marketing analyst benchmarking a cosmetics brand against competitors	Understand how my brand's pricing and ratings compare to the wider market	I lack a consolidated view of competitor pricing, ratings, and skin-type coverage	Market data is fragmented and not visualized in a way that supports quick benchmarking	Frustrated and unable to confidently justify pricing or positioning decisions with data

Key Takeaway

All three personas are blocked by the same root cause: cosmetics product data exists (price, rating, ingredients, skin-type suitability) but is not accessible or visual. Priya cannot compare price against rating before buying, Meera cannot see pricing and rating patterns across her catalog quickly enough to plan stock, and Arjun cannot benchmark his brand against competitors without manually cross-referencing spreadsheets. This shared root cause justified a single Tableau solution serving all three audiences — one shared, cleaned dataset with an interactive dashboard layered on top, rather than three separate reports.